

ATS Resume Scoring System – Detailed Project Report

Project Overview

The ATS Resume Scoring System is a Python-based application designed to simulate how real Applicant Tracking Systems (ATS) work in the recruitment process. The system analyzes resume documents and compares them with job descriptions to calculate a compatibility score using Natural Language Processing (NLP) and semantic similarity techniques.

Objectives

- 1 Automate the initial resume screening process
- 2 Compare resume content with job description requirements
- 3 Generate an ATS compatibility score (0–100)
- 4 Identify matched and missing skills
- 5 Provide actionable resume improvement suggestions
- 6 Help candidates improve resume quality for ATS systems

Technologies Used

- 1 Python 3.9–3.11 – Core programming language
- 2 Streamlit – Web-based user interface
- 3 spaCy – Natural Language Processing and tokenization
- 4 NLTK – Stopword removal and text normalization
- 5 Sentence-Transformers – Semantic text embeddings
- 6 Cosine Similarity – Resume and job description matching
- 7 PyPDF2 – Resume PDF text extraction

System Workflow

- 1 User uploads one or more resume PDF files
- 2 User pastes the job description into the application
- 3 Text is extracted from resume PDFs
- 4 NLP preprocessing removes stopwords and irrelevant tokens

- 5 Skills are extracted and compared with job description skills
- 6 Sentence embeddings are generated for resume and job description
- 7 Cosine similarity is calculated to determine ATS score
- 8 Section-wise analysis is performed
- 9 Results and suggestions are displayed in the UI
- 10 User downloads ATS analysis report

Step-by-Step Execution Process

- 1 Install Python 3.11 and ensure it is added to the system PATH.
- 2 Create a project folder named Resume_Scoring_System.
- 3 Open the folder in Visual Studio Code.
- 4 Save the application code as Resume_Scoring_System.py.
- 5 Trust the workspace when prompted by VS Code.
- 6 Install all required libraries using pip.
- 7 Download the spaCy English language model.
- 8 Navigate to the project directory using the terminal.
- 9 Run the Streamlit application using the streamlit run command.
- 10 Access the application in a web browser at localhost:8501.

Final Output

- 1 ATS score displayed on a scale of 0–100
- 2 List of matched and missing skills
- 3 Section-wise similarity scores
- 4 Resume improvement suggestions
- 5 Downloadable ATS analysis report

Advantages

- 1 Reduces manual effort in resume screening
- 2 Improves candidate awareness of ATS requirements
- 3 Provides structured and actionable feedback

- 4 Beginner-friendly NLP project
- 5 Cost-effective and easy to deploy

Future Enhancements

- 1 Integration with online job portals
- 2 Advanced skill ontology and database
- 3 Machine learning-based resume ranking
- 4 Cloud deployment using Streamlit Cloud or AWS
- 5 AI-powered resume rewriting suggestions

ATS Resume Scoring System – Process Flow Diagram

